

Functions of many variables – 3

Maps of $\mathbb{R}^n$ into itself: the Jacobian, invertibility and rigidity

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Theoretical minimum.

Theorem 1 (inverse function theorem). If f is C^1 near a and $\det Df(a) \neq 0$, then f is a diffeomorphism of a neighbourhood of a onto a neighbourhood of $f(a)$, and $D(f^{-1})(f(a)) = (Df(a))^{-1}$. The proof is the contraction principle applied to $x \mapsto x + Df(a)^{-1}(y - f(x))$. The statement is local and stays local: the map $(x, y) \mapsto (e^x \cos y, e^x \sin y)$ has non-vanishing Jacobian everywhere and is not injective.

Theorem 2 (implicit function theorem). If $F(x, y)$ is C^1 , $F(a, b) = 0$ and $\partial_y F(a, b)$ is invertible, then near (a, b) the equation $F(x, y) = 0$ defines a unique C^1 function $y = \varphi(x)$, and $\varphi' = -(\partial_y F)^{-1} \partial_x F$. It is the inverse function theorem applied to $(x, y) \mapsto (x, F(x, y))$.

Theorem 3 (Hadamard: from local to global). A local diffeomorphism $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ which is proper (preimages of compact sets are compact — for instance if $\|f(x)\| \rightarrow \infty$) is a diffeomorphism of $\mathbb{R}^n$ onto $\mathbb{R}^n$. Properness cannot be dropped, as the exponential example above shows.

Fact 1 (monotonicity gives injectivity). If the symmetric part $\frac{1}{2}(Df(x) + Df(x)^T)$ is positive definite everywhere, then $\langle f(b) - f(a), b - a \rangle > 0$ for $a \neq b$ and f is injective; one integrates the derivative along the segment. This is exactly what the determinant condition of problem 7 in the homework of Functions of many variables – 1 says for $n = 2$.

Fact 2 (rigidity). A map that carries a whole family of figures to figures of the same kind is usually affine. Two standard mechanisms: midpoints are definable from the family (so the map is additive up to a translation), and

continuity turns an additive map into a linear one. For distances, a map preserving one distance in $\mathbb{R}^n$, $n \geq 2$, already preserves all of them.

1. Prove the inverse function theorem from the contraction principle. Then show that $(x, y) \mapsto (e^x \cos y, e^x \sin y)$ is a local diffeomorphism at every point, is not injective, and that its image is $\mathbb{R}^2 \setminus \{0\}$; explain which hypothesis of Theorem 3 fails.
2. Prove Fact 1: if the symmetric part of Df is positive definite everywhere on a convex domain, then f is injective. Deduce the Putnam 2018 criterion for $n = 2$: positivity of $\frac{\partial f_1}{\partial x_1} \frac{\partial f_2}{\partial x_2} - \frac{1}{4} \left(\frac{\partial f_1}{\partial x_2} + \frac{\partial f_2}{\partial x_1} \right)^2$ is exactly positive definiteness of that symmetric part.
3. (Mexmat, 2017) Suppose we are given two maps $F, G: \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $\|F(x) - F(y)\| \leq \|x - y\|/2$, and G is continuous and has the property that the image under G of every closed ball contains the ball of the same radius centred at the image of the centre. Prove that there exists a point p with $F(p) = G(p)$.
4. (Proposed problems) Let $n \geq 1$ and let $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a map such that the image of every sphere of codimension 1 is a sphere of codimension 1 of the same radius. Prove that f is an isometry.
5. (ICMC 2023) A continuous function $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ maps circles to circles (a point is not a circle). Prove that it maps lines to lines.